BIO 101 — Final Exam — Spring 2023
Dr. Sarah — Total: 40 marks

Section A: MCQ (1 mark each)

1. Which of the following is an example of a prokaryotic cell?

   a. Animal cell
   b. Plant cell
   c. Bacteria
   d. Fungal cell

2. Which of the following modifies and packages proteins?

   a. Rough ER
   b. Golgi apparatus
   c. Ribosomes
   d. Mitochondria

3. Which of the following does NOT require energy?

   a. Na^+/K^+ pump
   b. Active transport
   c. Diffusion
   d. All of the above

4. Pepsin works best at a pH of about:

   a. 2
   b. 7
   c. 9
   d. 12

5. Which of the following happens to an enzyme above its optimum temperature?

   a. Its activity keeps increasing
   b. It becomes denatured
   c. It is consumed
   d. It binds more substrate

Section B: True / False

6. The plasma membrane is made of a phospholipid bilayer with proteins. ( )

7. Non-competitive inhibitors bind to the active site. ( )

Section C: Short questions

8. List the two types of transport across the membrane with one example of each. (4 marks)

9. Explain the factors that affect enzyme activity. (8 marks)